

Manas Singh

+91 8318916361 | manas.singh0503@gmail.com | msd25027@iiitl.ac.in | LinkedIn | GitHub

Professional Experience

- Data Science Intern – Climate Resilience Observatory (CRO), U.P. Govt.** Sept 2025 – Mar 2026
Machine Learning Modelling and Research *On-site*
- Developed and maintained a Machine Learning pipeline for rainfall prediction using real-world climatological data, achieving high accuracy across test splits.
 - Sourced and processed datasets from NASA POWER and IMD with focus on Uttar Pradesh regional climate patterns.
 - Contributed to multiple ongoing projects targeting climate risk assessment and agricultural planning support.
- Climate Data Science Intern – Darukaa.earth** Mar 2026 – June 2026
Climate Data Analysis and Modelling *Hybrid*
- Working on climate data pipelines and analysis focused on environmental sustainability and carbon-related datasets.
 - Applying machine learning techniques to derive actionable insights from ecological and climate data.
 - Converted Damage to the asset into financial loss to the company.

Education

- Indian Institute of Information Technology Lucknow** Aug 2025 – Jun 2027
M.Sc in Data Science, CGPA: 8.05 *Lucknow, India*
- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Reinforcement Learning, DBMS, Data Structures & Algorithms, Big Data Analytics, Statistics.
- University of Lucknow** Aug 2022 – Jun 2025
B.Sc in Statistics *Lucknow, India*
- Relevant Coursework: Linear Algebra, Probability & Statistics, Time Series, Hypothesis Testing, Operations Research.

Technical Projects

- FinSightOS – AI Investment Research Platform** GitHub
Python, Agents, MCP, FineTuning, RAG, Yfinance, Redis, Docker, PostgreSQL
- Building FinSightOS – a 7-agent AI investment research platform using LangGraph, 6 MCP servers, and RAG on live NSE/BSE data; implements industry-standard DCF, RSI, Sharpe ratio, and VaR calculations following CFA-standard formulas.
 - Using Groq & Fine-tuned Mistral-7B model on the FinQA Dataset for financial question answering.
 - Giving complete report and analysis on Indian listed companies at one step platform.
- Regulatory Change Autopilot – AI-Powered Compliance Intelligence System** GitHub
Python, HuggingFace, MCP, LLM, Qdrant, RAG
- Built a live RAG pipeline scraping RBI/SEBI/IRDAI circulars, extracting PDF text, embedding with HuggingFace, and storing in Qdrant for semantic retrieval.
 - Exposed the full pipeline as MCP tools, enabling AI agents to autonomously detect circulars, analyse gaps, and generate reports.
 - Reduced the manual analysis time from 3-4 days to minutes using AI agents.
- Plant Disease Classification System & AI Chatbot** GitHub
Python, FastAPI, CNN, GCP, Matplotlib, TensorFlow
- Developed a CNN-based DL model for plant disease classification, achieving 95% accuracy across 3 disease categories.
 - Built and deployed on Render a FastAPI backend for real-time image inference and HTML, CSS, JS for image upload and diagnosis visualization.
 - Integrated Groq LLM powered chatbot using Langchain for conversational Response.

Technical Expertise

Programming & Tools: Python, SQL, C, Git, Data Structures & Algorithms, Docker, VectorDB
Machine Learning & Deep Learning: Supervised & Unsupervised Learning, ANN, CNN, Attention Transformers, LLMs, NLP
Generative AI & RAG: LangChain, LangGraph, HuggingFace Transformers, Agentic Systems, MCP, RAG, Fine-Tuning (LoRA/QLoRA)
Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn

Achievements

- Qualified IIT JAM Mathematical Statistics with AIR 202.
- Solved 100+ DSA and SQL problems on LeetCode.
- Submitted a research paper at INGARSS Conference 2026 on Rainfall Prediction.